

Chanyoung Park

CONTACT INFORMATION

Associate Professor
Data Science & Artificial Intelligence Lab (DSAIL)
Dept. of Industrial & Systems Engineering, KAIST
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Graduate School of Data Science, KAIST
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[Google Scholar](#)

CURRENT RESEARCH

- **AI for Scientific Discovery**
 - Materials, Chemistry
 - Biology, Engineering, EDA
- **Agentic AI**
- **LLM / Multimodal Data Mining**
 - Recommender System, Graph Neural Networks
 - LLM Safety, Model Efficiency, Video Language Model
 - Scene Graph Generation, Anomaly Detection, Time Series Analysis
- **Data-Centric AI**
 - Model Robustness, Adversarial Attack/Defense
 - Imbalanced Learning, Few-shot Learning, Explainability

POSITIONS

- KAIST**, Daejeon, South Korea Mar 2025 – Present
- Associate Professor, Dept. of Industrial & Systems Engineering
 - Associate Professor, The Kim Jaechul Graduate School of AI
 - Associate Professor, Graduate School of Data Science
- KAIST**, Daejeon, South Korea Nov 2020 – Feb 2025
- Assistant Professor, Dept. of Industrial & Systems Engineering
 - Assistant Professor, The Kim Jaechul Graduate School of AI
 - Assistant Professor, Graduate School of Data Science
- University of Illinois at Urbana-Champaign**, Urbana, IL, USA Jan 2019 – Oct 2020
- Postdoctoral Reserach Fellow in Computer Science Department
 - Host: [Prof. Jiawei Han](#)
 - Topics: Multimodal data mining, Heterogeneous network representation learning
- Microsoft Research**, Beijing, China Sep 2017 – Dec 2017
- Research Intern
 - Mentor: [Dr. Xing Xie](#)
 - Project: Collaborative Translational Metric Learning
- NAVER**, Seongnam, South Korea Mar 2017 – Aug 2017
- Research Intern
 - Mentors: [Dr. Jung-Tae Lee](#), and [Dr. Min-Chul Yang](#)
 - Project: Click-aware Purchase Prediction with Push at the Top

EDUCATION

- Pohang University of Science and Technology**, Pohang, South Korea Feb 2019
- Ph.D. in Computer Science and Engineering
 - Thesis: Recommendation Framework via Matrix Factorization and Translation
 - Topics: Multimodal data mining, User behavior modeling, Recommender system
- Sogang University**, Seoul, South Korea Feb 2014
- B.S. in Computer Science and Engineering

AWARDS & SCHOLARSHIPS

- Best Paper Award** 2026
- KDD 2026 Workshop on Personal Intelligence in Agentic AI Era, Jeju, South Korea
- Best Paper Award** 2026
- KDD 2026 Workshop on Reliable Scientific Foundation Models, Jeju, South Korea
- Google Research Award** 2026

▪ Google	
Impact Research Award (Achievement Award)	2025
▪ KAIST	
Best Academic Award	2025
▪ Dept. of Industrial and Systems Engineering, KAIST	
Song-Am Distinguished Research Award	2024
▪ College of Engineering, KAIST	
Best Academic Award	2024
▪ Dept. of Industrial and Systems Engineering, KAIST	
Best Paper Award	2024
▪ KDD 2024 Workshop on Federated Learning for Data Mining and Graph Analytics (FedKDD), Barcelona, Spain	
Best Paper Award	2024
▪ KDD 2024 Workshop on Human-Interpretable AI, Barcelona, Spain	
Gold Prize, 30th Samsung Humantech Paper Award	2024
Technology Innovation Awards	2023
▪ College of Engineering, KAIST	
Top Reviewer of NeurIPS	2023
Best Paper Award	2023
▪ ICML 2023 Workshop on Computational Biology, Hawaii, USA	
KAIST Excellence in Teaching Award of Year 2022	2023
Best Teaching Award	2023
▪ Dept. of Industrial and Systems Engineering, KAIST	
ICDM Travel Award	2018
▪ International Conference on Data Mining, Singapore	
Award of excellence	2017
▪ Microsoft Research Asia Internship Program, Beijing, China	
▪ Awarded to the best intern	
Naver Ph.D Fellowship	2016
▪ Awarded to outstanding Ph.D students majoring in computer science in Korea	
RecSys Challenge 2015	2015
▪ Ranked 10 out of 850 teams from 49 countries (Top 1.1%)	
▪ Task: Click/Purchase Prediction	
Dean's List	Fall 2013
▪ Sogang University	

PUBLICATIONS

CONFERENCES

- [C92] Compositional Spectral Prompts for LLM-based Online Time Series Forecasting
Seungyeon Choi, Hyunchul Kim, Jae-Gil Lee, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management (**CIKM 2026**)
- [C91] ELASTIC: Trajectory-Based Synchronization of Event and Tracking Data in Soccer
Hyunsung Kim, Hoyoung Choi, Kunhee Lee, Sangwoo Seo, Tom Boomstra, Jinsung Yoon,
Chanyoung Park
ACM International Conference on Information and Knowledge Management (**CIKM 2026**)
- [C90] DefaultGNN: A Dual-Perspective GNN Framework for Predicting Corporate Default from
Buyer-Seller Transaction Networks
Junghoon Kim, Hyunsung Kim, Seungyeon Choi, Kyoungyong Park, Jihun Lee, Yonggu Ji,
Chanyoung Park
ACM International Conference on Information and Knowledge Management (**CIKM 2026**)
(**Applied Research Track**)
- [C89] A Behavioral Trait Leaks into Preferences: Diagnosing Trait Interference in LLM User Simulators
Chaehyun Kim, Sein Kim, Hongseok Kang, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management (**CIKM 2026**) (**Short Paper**)
- [C88] HUGETIME at eRisk 2026: Bucket-wise Evidence Pruning for Contextualized Early Depression
Detection
Ilho Shin, Hyunchul Kim, **Chanyoung Park**
The Conference and Labs of the Evaluation Forum (**CLEF 2026**)
Ranked 1st in Task 2: Contextualized Early Detection of Depression at CLEF eRisk 2026

- [C87] Sink-Token-Aware Pruning for Fine-Grained Video Understanding in Efficient Video LLMs
Kibum Kim, Jiwan Kim, Kyle Min, Yueqi Wang, Jinyoung Moon, Julian McAuley, **Chanyoung Park**
European Conference on Computer Vision (**ECCV 2026**)
- [C86] Why and When Visual Token Pruning Fails? A Study on Relevant Visual Information Shift in MLLMs Decoding
Jiwan Kim, Kibum Kim, Wonjoong Kim, Byung-Kwan Lee, **Chanyoung Park**
European Conference on Computer Vision (**ECCV 2026**)
- [C85] PathCRF: Ball-Free Soccer Event Detection via Possession Path Inference from Player Trajectories
Hyunsung Kim, Kunhee Lee, Sangwoo Seo, Sang-Ki Ko, Jinsung Yoon, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2026**)
- [C84] Learning Probabilistic Compositional Representation of Crystalline Materials
Namkyeong Lee, Heewoong Noh, Gyoung S. Na, Jimeng Sun, Tianfan Fu, Marinka Zitnik, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2026**) (**AI for Science Track**)
- [C83] EqGINO: Equivariant Geometry-Informed Fourier Neural Operators for 3D PDEs
Sungwon Kim, Juho Song, Seungmin Shin, Guimok Cho, Sangkook Kim, **Chanyoung Park**
International Conference on Machine Learning (**ICML 2026**)
- [C82] Beyond the Final Answer: Evaluating the Reasoning Trajectories of Tool-Augmented Agents
Wonjoong Kim, Sangwu Park, Yeonjun In, Sein Kim, Dongha Lee, **Chanyoung Park**
International Conference on Machine Learning (**ICML 2026**)
- [C81] Learning Adaptive Perturbation-Conditioned Contexts for Robust Transcriptional Response Prediction
Yinhua Piao, Hyomin Kim, Seonghwan Kim, Yunhak Oh, Junhyeok Jeon, Sang-Yeon Hwang, Jaechang Lim, Woo Youn Kim, **Chanyoung Park**, Sungsoo Ahn
International Conference on Machine Learning (**ICML 2026**)
- [C80] Time-PEFT: Temporal and Multichannel Complexity-Based Fine-Tuning for Time-Series Foundation Models
Jihye Na, Patara Trirat, **Chanyoung Park**, Jae-Gil Lee
International Conference on Machine Learning (**ICML 2026**)
- [C79] SelfJudge: Faster Speculative Decoding via Self-Supervised Judge Verification
Kanghoon Yoon, Minsub Kim, Sungjae Lee, Joonhyung Lee, Sunghyeon Woo, Yeonjun In, Se Jung Kwon, **Chanyoung Park**, Dongsoo Lee
International Conference on Machine Learning (**ICML 2026**)
- [C78] Reasoning Structure Matters for Safety Alignment of Reasoning Models
Yeonjun In, Wonjoong Kim, Sangwu Park, **Chanyoung Park**
The 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026**)
- [C77] IR-Agent: Expert-Inspired LLM Agents for Structure Elucidation from Infrared Spectra
Heewoong Noh, Namkyeong Lee, Gyoung S. Na, Kibum Kim, **Chanyoung Park**
The Fourteenth International Conference on Learning Representations (**ICLR 2026**)
- [C76] CompoDistill: Attention Distillation for Compositional Reasoning in Multimodal LLMs
Jiwan Kim, Kibum Kim, Sangwoo Seo, **Chanyoung Park**
The Fourteenth International Conference on Learning Representations (**ICLR 2026**)
- [C75] Token-Efficient Item Representation via Images for LLM Recommender Systems
Kibum Kim, Sein Kim, HongSeok Kang, Jiwan Kim, Heewoong Noh, Yeonjun In, Kanghoon Yoon, Jinoh Oh, Julian McAuley, **Chanyoung Park**
The Fourteenth International Conference on Learning Representations (**ICLR 2026**)
- [C74] PULSE: Socially-Aware User Representation Modeling Toward Parameter-Efficient Graph Collaborative Filtering
Doyun Choi*, Cheonwoo Lee*, Biniyam Aschalew Tolera, Taewook Ham, **Chanyoung Park**, Jaemin Yoo
The Web Conference (**WWW 2026**)

- [C73] RAG-Enhanced Collaborative LLM Agents for Drug Discovery
Namkyeong Lee, Edward De Brouwer, Ehsan Hajiramezanali, Tommaso Biancalani, **Chanyoung Park**, Gabriele Scalia
The 40th Annual AAAI Conference on Artificial Intelligence (**AAAI 2026**)
- [C72] 3D Interaction Geometric Pre-training for Molecular Relational Learning
Namkyeong Lee, Yunhak Oh, Heewoong Noh, Gyoung S. Na, Minkai Xu, Hanchen Wang, Tianfan Fu, **Chanyoung Park**
The Thirty-Ninth Annual Conference on Neural Information Processing Systems (**NeurIPS 2025 (Spotlight)**)
- [C71] Disentangling Hyperedges through the Lens of Category Theory
Yoonho Lee, Junseok Lee, Sangwoo Seo, Sungwon Kim, Yeongmin Kim, **Chanyoung Park**
The Thirty-Ninth Annual Conference on Neural Information Processing Systems (**NeurIPS 2025**)
- [C70] Training Robust Graph Neural Networks by Modeling Noise Dependencies
Yeonjun In, Kanghoon Yoon, Sukwon Yun, Kibum Kim, Sungchul Kim, **Chanyoung Park**
The Thirty-Ninth Annual Conference on Neural Information Processing Systems (**NeurIPS 2025**)
- [C69] Is Safety Standard Same for Everyone? User-Specific Safety Evaluation of Large Language Models
Yeonjun In, Wonjoong Kim, Kanghoon Yoon, Sungchul Kim, Mehrab Tanjim, Kibum Kim, **Chanyoung Park**
Conference on Empirical Methods in Natural Language Processing (**EMNLP 2025 Findings**)
- [C68] Disambiguation in Conversational Question Answering in the Era of LLM: A Survey
Md Mehrab Tanjim*, Yeonjun In*, Xiang Chen, Victor S. Bursztyn, Ryan A. Rossi, Sungchul Kim, Guang-Jie Ren, Vaishnavi Muppala, Shun Jiang, Yongsung Kim, **Chanyoung Park**
Conference on Empirical Methods in Natural Language Processing (**EMNLP 2025**)
- [C67] Target Circuit Matching in Large-Scale Netlists using GNN-Based Region Prediction
Sangwoo Seo, Jimin Seo, Yoonho Lee, Donghyeon Kim, Hyejin Shin, Banghyun Sung, **Chanyoung Park**
International Conference on Computer-Aided Design (**ICCAD 2025**)
- [C66] Lost in Sequence: Do Large Language Models Understand Sequential Recommendation?
Sein Kim*, HongSeok Kang*, Kibum Kim, Jiwan Kim, Donghyun Kim, Min-Chul Yang, Kwangjin Oh, Julian McAuley, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2025**)
- [C65] Oldie but Goodie: Re-illuminating Label Propagation on Graphs with Partially Observed Features
Sukwon Yun, Xin Liu, Yunhak Oh, Junseok Lee, Tianlong Chen, Tsuyoshi Murata, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2025**)
- [C64] Electron-Informed Coarse-Graining Molecular Representation Learning for Real-World Molecular Physics
Gyoung S. Na, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2025**)
- [C63] On Measuring Unnoticeability of Graph Adversarial Attacks: Observations, New Measure, and Applications
Hyeonsoo Jo*, Hyunjin Hwang*, Fanchen Bu, Soo Yong Lee, **Chanyoung Park**, Kijung Shin
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2025**)
- [C62] Thickness-aware E(3)-Equivariant 3D Mesh Neural Networks
Sungwon Kim, Namkyeong Lee, Yunyoung Doh, Seungmin Shin, Guimok Cho, Seung-Won Jeon, Sangkook Kim, **Chanyoung Park**
International Conference on Machine Learning (**ICML 2025**)
- [C61] Global Context-aware Representation Learning for Spatially Resolved Transcriptomics
Yunhak Oh*, Junseok Lee*, Yeongmin Kim, Sangwoo Seo, Namkyeong Lee, **Chanyoung Park**
International Conference on Machine Learning (**ICML 2025**)
- [C60] Disentangling and Generating Modalities for Recommendation in Missing Modality Scenarios
Jiwan Kim, Hongseok Kang, Sein Kim, Kibum Kim, **Chanyoung Park**
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2025**)

- [C59] Dynamic Time-aware Continual User Representation Learning
Seungyoon Choi, Sein Kim, Hongseok Kang, Wonjoong Kim, **Chanyoung Park**
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2025**)
- [C58] Subgraph Federated Learning for Local Generalization
Sungwon Kim, Yoonho Lee, Yunhak Oh, Namkyeong Lee, Sukwon Yun, Junseok Lee, Sein Kim, Carl Yang, **Chanyoung Park**
The Thirteenth International Conference on Learning Representations (**ICLR 2025 (Oral)**)
- [C57] Weakly Supervised Video Scene Graph Generation via Natural Language Supervision
Kibum Kim, Kanghoon Yoon, Yeonjun In, Jaehyeong Jeon, Jinyoung Moon, Donghyun Kim, **Chanyoung Park**
The Thirteenth International Conference on Learning Representations (**ICLR 2025**)
- [C56] Self-Supervised Diffusion Models for Electron-Aware Molecular Representation Learning
Gyoung S. Na, **Chanyoung Park**
The Thirteenth International Conference on Learning Representations (**ICLR 2025**)
- [C55] Diversify-verify-adapt: Efficient and Robust Retrieval-Augmented Ambiguous Question Answering
Yeonjun In, Sungchul Kim, Ryan A. Rossi, Mehrab Tanjim, Tong Yu, Ritwik Sinha, **Chanyoung Park**
Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (**NAACL 2025**)
- [C54] SIMPLOT: Enhancing Chart Question Answering by Distilling Essentials
Wonjoong Kim*, Sangwu Park*, Yeonjun In, Seokwon Han, **Chanyoung Park**
Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (**NAACL 2025 Findings**)
- [C53] RA-SGG: Retrieval-Augmented Scene Graph Generation Framework via Multi-Prototype Learning
Kanghoon Yoon, Kibum Kim, Jaehyeong Jeon, Yeonjun In, Donghyun Kim, **Chanyoung Park**
Thirty-Ninth AAAI Conference on Artificial Intelligence (**AAAI 2025**)
- [C52] Revisiting Fake News Detection: Towards Temporality-aware Evaluation by Leveraging Engagement Earliness
Junghoon Kim*, Junmo Lee*, Yeonjun In, Kanghoon Yoon, **Chanyoung Park**
The 18th ACM International Conference on Web Search and Data Mining (**WSDM 2025 (Oral)**)
- [C51] Retrieval-Retro: Retrieval-based Inorganic Retrosynthesis with Expert Knowledge
Heewoong Noh, Namkyeong Lee, Gyoung S. Na, **Chanyoung Park**
Thirty-eighth Conference on Neural Information Processing Systems (**NeurIPS 2024**)
- [C50] Debaised Graph Poisoning Attack via Contrastive Surrogate Objective
Kanghoon Yoon, Yeonjun In, Namkyeong Lee, Kibum Kim, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management (**CIKM 2024**)
- [C49] Vision Language Model is NOT All You Need: Augmentation Strategies for Molecule Language Models
Namkyeong Lee, Siddhartha Laghuvarapu, **Chanyoung Park**, Jimeng Sun
ACM International Conference on Information and Knowledge Management (**CIKM 2024**)
- [C48] Semantic Diversity-aware Prototype-based Learning for Unbiased Scene Graph Generation
Jaehyeong Jeon, Kibum Kim, Kanghoon Yoon, **Chanyoung Park**
European Conference on Computer Vision (**ECCV 2024**)
- [C47] Multiplexed Immunofluorescence Image Analysis through an Efficient Multiplex Network
Sukwon Yun, Jie Peng, Alexandro Trevino, **Chanyoung Park**, Tianlong Chen
European Conference on Computer Vision (**ECCV 2024**)
- [C46] Large Language Models meet Collaborative Filtering: An Efficient All-round LLM-based Recommender System
Sein Kim*, Hongseok Kang*, Seungyoon Choi, Donghyun Kim, Minchul Yang, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2024**)
- [C45] Self-Explainable Temporal Graph Networks based on Graph Information Bottleneck
Sangwoo Seo, Sungwon Kim, Jihyeong Jung, Yoonho Lee, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2024**)

- [C44] Unsupervised Episode Generation for Graph Meta-learning
Jihyeong Jung, Sangwoo Seo, Sungwon Kim, **Chanyoung Park**
International Conference on Machine Learning (**ICML 2024**)
- [C43] LLM4SGG: Large Language Models for Weakly Supervised Scene Graph Generation
Kibum Kim, Kanghoon Yoon, Jaehyeong Jeon, Yeonjun In, Jinyoung Moon, Donghyun Kim, **Chanyoung Park**
IEEE/CVF Computer Vision and Pattern Recognition Conference (**CVPR 2024**)
- [C42] Self-guided Robust Graph Structure Refinement
Yeonjun In, Kanghoon Yoon, Kibum Kim, Kijung Shin, **Chanyoung Park**
The Web Conference 2024 (**WWW 2024 (Oral)**)
- [C41] DSLR: Diversity Enhancement and Structure Learning for Rehearsal-based Graph Continual Learning
Seungyeon Choi*, Wonjoong Kim*, Sungwon Kim, Yeonjun In, Sein Kim, **Chanyoung Park**
The Web Conference 2024 (**WWW 2024 (Oral)**)
- [C40] Adaptive Self-training Framework for Fine-grained Scene Graph Generation
Kibum Kim*, Kanghoon Yoon*, Yeonjun In, Jinyoung Moon, Donghyun Kim, **Chanyoung Park**
Twelfth International Conference on Learning Representations (**ICLR 2024**)
- [C39] Sterling: Synergistic Representation Learning on Bipartite Graphs
Baoyu Jing, Yuchen Yan, Kaize Ding, **Chanyoung Park**, Yada Zhu, Huan Liu, Hanghang Tong
Thirty-Eighth AAAI Conference on Artificial Intelligence (**AAAI 2024 (Oral)**)
- [C38] Interpretable Prototype-based Graph Information Bottleneck
Sangwoo Seo, Sungwon Kim, **Chanyoung Park**
Thirty-seventh Conference on Neural Information Processing Systems (**NeurIPS 2023**) (**Gold Prize, 30th Samsung Humantech Paper Award**)
- [C37] Density of States Prediction of Crystalline Materials via Prompt-guided Multi-Modal Transformer
Namkyeong Lee*, Heewoong Noh*, Sungwon Kim, Dongmin Hyun, Gyoung S. Na, **Chanyoung Park**
Thirty-seventh Conference on Neural Information Processing Systems (**NeurIPS 2023**)
- [C36] MUSE: Music Recommender System with Shuffle Play Recommendation Enhancement
Yunhak Oh*, Sukwon Yun*, Dongmin Hyun, Sein Kim, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management (**CIKM 2023**)
- [C35] Class Label-aware Graph Anomaly Detection
Junghoon Kim, Yeonjun In, Kanghoon Yoon, Junmo Lee, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management (**CIKM 2023**) (Short Paper)
- [C34] Structural Mixup for Graph Neural Networks
Junghun Kim*, Sukwon Yun*, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management (**CIKM 2023**) (Short Paper)
- [C33] Similarity Preserving Adversarial Graph Contrastive Learning
Yeonjun In*, Kanghoon Yoon*, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)
- [C32] Shift-Robust Molecular Relational Learning with Causal Substructure
Namkyeong Lee, Kanghoon Yoon, Gyoung S. Na, Sein Kim, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)
- [C31] Task-Equivariant Graph Few-shot Learning
Sungwon Kim, Junseok Lee, Namkyeong Lee, Wonjoong Kim, Seungyeon Choi, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)
- [C30] Task Relation-aware Continual User Representation Learning
Sein Kim, Namkyeong Lee, Donghyun Kim, Min-Chul Yang, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)
- [C29] Conditional Graph Information Bottleneck for Molecular Relational Learning
Namkyeong Lee, Dongmin Hyun, Gyoung S. Na, Sungwon Kim, Junseok Lee, **Chanyoung Park**
International Conference on Machine Learning (**ICML 2023**)

- [C28] MELT: Mutual Enhancement of Long-Tailed User and Item for Sequential Recommendation
Kibum Kim, Dongmin Hyun, Sukwon Yun, **Chanyoung Park**
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2023**)
- [C27] Unbiased Heterogeneous Scene Graph Generation with Relation-aware Message Passing Neural Network
Kanghoon Yoon*, Kibum Kim*, Jinyoung Moon, **Chanyoung Park**
Thirty-Seventh AAAI Conference on Artificial Intelligence (**AAAI 2023**)
- [C26] Heterogeneous Graph Learning for Multi-modal Medical Data Analysis
Sein Kim, Namkyeong Lee, Junseok Lee, Dongmin Hyun, **Chanyoung Park**
Thirty-Seventh AAAI Conference on Artificial Intelligence (**AAAI 2023 (Oral)**)
- [C25] Generating Multiple-Length Summaries via Reinforcement Learning for Unsupervised Sentence Summarization
Dongmin Hyun, Xiting Wang, **Chanyoung Park**, Xing Xie, Hwanjo Yu
Conference on Empirical Methods in Natural Language Processing (**EMNLP 2022 Findings**)
- [C24] Set2Box: Similarity Preserving Representation Learning for Sets
Geon Lee, **Chanyoung Park**, Kijung Shin
The IEEE International Conference on Data Mining (**ICDM 2022**)
- [C23] LTE4G: Long-Tail Experts for Graph Neural Networks
Sukwon Yun, Kibum Kim, Kanghoon Yoon, **Chanyoung Park**
ACM International Conference on Information and Knowledge (**CIKM 2022**)
- [C22] Relational Self-Supervised Learning on Graphs
Namkyeong Lee, Dongmin Hyun, Junseok Lee, **Chanyoung Park**
ACM International Conference on Information and Knowledge (**CIKM 2022**)
- [C21] Beyond Learning from Next Item: Sequential Recommendation via Personalized Interest Sustainability
Dongmin Hyun, **Chanyoung Park**, Junsu Cho, Hwanjo Yu
ACM International Conference on Information and Knowledge (**CIKM 2022**)
- [C20] Nonlinearity Encoding for Extrapolation of Neural Networks
Gyoung S. Na, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2022**)
- [C19] GraFN: Semi-Supervised Node Classification on Graph with Few Labels via Non-Parametric Distribution Assignment
Junseok Lee, Yunhak Oh, Yeonjun In, Namkyeong Lee, Dongmin Hyun, **Chanyoung Park**
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2022**) (Short Paper)
- [C18] AHP: Learning to Negative Sample for Hyperedge Prediction
Hyunjin Hwang, Seungwoo Lee, **Chanyoung Park**, Kijung Shin
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2022**) (Short Paper)
- [C17] Augmentation-Free Self-Supervised Learning on Graphs
Namkyeong Lee, Junseok Lee, **Chanyoung Park**
Thirty-Sixth AAAI Conference on Artificial Intelligence (**AAAI 2022**)
- [C16] Learnable Structural Semantic Readout for Graph Classification
Dongha Lee, Su Kim, Seonghyeon Lee, **Chanyoung Park**, Hwanjo Yu
IEEE International Conference on Data Mining (**ICDM 2021**) (Short Paper)
- [C15] Bootstrapping User and Item Representations for One-Class Collaborative Filtering
Dongha Lee, Seongku Kang, Hyunjun Ju, **Chanyoung Park**, Hwanjo Yu
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2021**)
- [C14] HDMI: High-order Deep Multiplex Infomax
Baoyu Jing, **Chanyoung Park**, Hanghang Tong
The Web Conference (**WWW 2021**)

- [C13] Interest Sustainability-Aware Recommender System
Dongmin Hyun, Junsu Cho, **Chanyoung Park**, Hwanjo Yu
IEEE International Conference on Data Mining (**ICDM 2020**)
- [C12] Unsupervised Differentiable Multi-aspect Network Embedding
Chanyoung Park, Carl Yang, Qi Zhu, Donghyun Kim, Hwanjo Yu and Jiawei Han
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2020**)
- [C11] Unsupervised Attributed Multiplex Network Embedding
Chanyoung Park, Donghyun Kim, Jiawei Han, Hwanjo Yu
Thirty-Fourth AAAI Conference on Artificial Intelligence (**AAAI 2020**)
- [C10] Task-Guided Pair Embedding in Heterogeneous Network
Chanyoung Park, Donghyun Kim, Qi Zhu, Jiawei Han, Hwanjo Yu
ACM International Conference on Information and Knowledge (**CIKM 2019**)
- [C9] BHIN2vec: Balancing the Type of Relation in Heterogeneous Information Network
Seonghyun Lee, **Chanyoung Park**, Hwanjo Yu
ACM International Conference on Information and Knowledge (**CIKM 2019**)
- [C8] Action Space Learning for Heterogeneous User Behavior Prediction
Dongha Lee, **Chanyoung Park**, Hyunjun Ju, Junyoung Hwang, Hwanjo Yu
International Joint Conference on Artificial Intelligence (**IJCAI 2019**)
- [C7] Sequential and Diverse Recommendation with Long Tail
Yejin Kim, Kwangseob Kim, **Chanyoung Park**, Hwanjo Yu
International Joint Conference on Artificial Intelligence (**IJCAI 2019**)
- [C6] Collaborative Translational Metric Learning
Chanyoung Park, Donghyun Kim, Xing Xie, Hwanjo Yu
IEEE International Conference on Data Mining (**ICDM 2018**)
- [C5] Review Sentiment-Guided Scalable Deep Recommender System
Dongmin Hyun, **Chanyoung Park**, Min-Chul Yang, Ilhyeon Song, Jung-Tae Lee, Hwanjo Yu
ACM SIGIR conference on Research and Development in Information Retrieval (**SIGIR 2018**)
(Short Paper)
- [C4] Do “Also-Viewed” Products Help User Rating Prediction?
Chanyoung Park, Donghyun Kim, Jinoh Oh, Hwanjo Yu
International World-Wide Web Conference (**WWW 2017**)
- [C3] Convolutional Matrix Factorization for Document Context-Aware Recommendation
Donghyun Kim, **Chanyoung Park**, Jinoh Oh, Hwanjo Yu
ACM international conference on Recommender System (**RecSys 2016**)
- [C2] TRecSo: Enhancing Top-k Recommendation With Social Information
Chanyoung Park, Donghyun Kim, Jinoh Oh, Hwanjo Yu
International World-Wide Web Conference (**WWW 2016**) (Poster)
- [C1] Predicting User Purchase in E-commerce by Comprehensive Feature Engineering and Decision Boundary Focused Under-Sampling
Chanyoung Park, Donghyun Kim, Jinoh Oh, Hwanjo Yu
ACM international conference on Recommender System Challenge (**ACM RecSysChallenge 2015**, 10th place, top 1.1%)

JOURNALS

- [J14] A New DAWN for Fake News Detection: Leveraging Engagement Earliness for Temporality-aware Evaluation
Junghoon Kim*, Junmo Lee*, Yeonjun In, Kanghoon Yoon, **Chanyoung Park**
ACM Transactions on Intelligent Systems and Technology (2026)
- [J13] Physics-Embedded Graph Neural Operator for Interaction-Controlled Colloidal Aggregation
Yongjoon Choe*, Sungwon Kim*, Susan E. Burns, **Chanyoung Park**
Water Research (2026)
- [J12] Single-cell RNA Sequencing Data Imputation Using Bi-level Feature Propagation
Junseok Lee*, Sukwon Yun*, Yeongmin Kim, Tianlong Chen, Manolis Kellis, **Chanyoung Park**
Briefings in Bioinformatics (2024)

- [J11] Deep Single-cell RNA-seq Data Clustering with Graph Prototypical Contrastive Learning
Junseok Lee, Sungwon Kim, Dongmin Hyun, Namkyeong Lee, Yejin Kim, **Chanyoung Park**
Bioinformatics (2023)
- [J10] Self-Supervised Graph Representation Learning via Positive Mining
Namkyeong Lee, Junseok Lee, **Chanyoung Park**
Information Sciences (2022.9)
- [J9] Learning to Utilize Auxiliary Reviews for Recommendation
Dongmin Hyun, **Chanyoung Park**, Junsu Cho, Hwanjo Yu
Information Sciences (2021.2)
- [J8] Deep Multiplex Graph Infomax: Attentive Multiplex Network Embedding using Global Information
Chanyoung Park, Jiawei Han, Hwanjo Yu
Knowledge-Based Systems (2020.6)
- [J7] Improved prediction of miRNA–disease associations based on matrix completion with network regularization
Jihwan Ha, Chihyun Park, **Chanyoung Park**, Sanghyun Park
Cells (2020.4) (SCIE)
- [J6] Click-aware Purchase Prediction with Push at the Top
Chanyoung Park, Donghyun Kim, Min-Chul Yang, Jung-Tae Lee, Hwanjo Yu
Information Sciences (2020.6)
- [J5] Inferring miRNA-disease Interactions using Probabilistic Matrix Factorization
Jihwan Ha, Chihyun Park, **Chanyoung Park**, Sanghyun Park
Journal of Biomedical Informatics (2020.2)
- [J4] An Encoder-Decoder Switch Network for Purchase Prediction
Chanyoung Park, Donghyun Kim, Hwanjo Yu
Knowledge-Based Systems (2019.12)
- [J3] Target-aware Convolutional Neural Network for Target-level Sentiment Analysis
Dongmin Hyun, **Chanyoung Park**, Min-Chul Yang, Ilhyeon Song, Jung-Tae Lee, Hwanjo Yu
Information Sciences (2019.7)
- [J2] Deep Hybrid Recommender Systems via Exploiting Document Context and Statistics of Items
Donghyun Kim, **Chanyoung Park**, Jinoh Oh, Hwanjo Yu
Information Sciences (2017.11)
- [J1] Improving top-K recommendation with truster and trustee relationship in user trust network
Chanyoung Park, Donghyun Kim, Jinoh Oh, Hwanjo Yu
Information Sciences (2016.12)

WORKSHOPS

- [W28] Tokenizing Antibody–Antigen Complex Structures via Energy-Based Surrogates
Junseok Lee, Yunhak Oh, Namkyeong Lee, Hyunchul Kim, **Chanyoung Park**
Workshop on Reliable Scientific Foundation Models (**KDD 2026 Workshop**)
- [W27] Bridging Circuit Graphs and Language: A Cross-Modal Transformer for Netlist-to-Text Alignment and Generation
Sangwoo Seo, Hyunsung Kim, Hyunchul Kim, Byeonggon Kang, Rex Ying, **Chanyoung Park**
Workshop on Frontiers in Graph Machine Learning for the Large Model Era (**KDD 2026 Workshop**)
- [W26] Adaptive Graph Rewiring to Mitigate Over-Squashing in Mesh-Based GNNs for Fluid Dynamics Simulations
Sangwoo Seo, Hyunsung Kim, Jiwan Kim, **Chanyoung Park**
Workshop on Reliable Scientific Foundation Models (**KDD 2026 Workshop**)
- [W25] A Structured LLM Framework for Inorganic Material Synthesis Planning
Heewoong Noh, Gyoung S. Na, Namkyeong Lee, **Chanyoung Park**
Workshop on Agentic AI for Scientific and Societal Advances (**KDD 2026 Workshop**)
- [W24] Beyond the Final Answer: Evaluating the Reasoning Trajectories of Tool-Augmented Agents
Wonjoong Kim, Sangwu Park, Yeonjun In, Sein Kim, Dongha Lee, **Chanyoung Park**
Workshop on Evaluation and Trustworthiness of Agentic AI (**KDD 2026 Workshop**)

- [W23] Compositional Spectral Prompts for LLM-based Online Time Series Forecasting
Seungyoon Choi, Hyunchul Kim, Jae-Gil Lee, **Chanyoung Park**
Workshop on Mining and Learning from Time Series (**KDD 2026 Workshop**)
- [W22] Personalize-then-Store: Benchmarking and Learning Personalized Memory for Long-Horizon Agents
Yeonjun In, Wonjoong Kim, Sangwu Park, Kanghoon Yoon, **Chanyoung Park**
Workshop on Personal Intelligence in Agentic AI Era (**KDD 2026 Workshop**)
- [W21] Rethinking Failure Attribution in Multi-Agent Systems: A Multi-Perspective Benchmark and Evaluation
Yeonjun In, Mehrab Tanjim, Jayakumar Subramanian, Sungchul Kim, Uttaran Bhattacharya, Wonjoong Kim, Sangwu Park, Somdeb Sarkhel, **Chanyoung Park**
Workshop on Agentic Software Engineering (**KDD 2026 Workshop**)
- [W20] Self-EvolveRec: Self-Evolving Recommender Systems with LLM-based Directional Feedback
Sein Kim, Sangwu Park, HongSeok Kang, Wonjoong Kim, Jimin Seo, Yeonjun In, Kanghoon Yoon, Hyunsik Jeon, **Chanyoung Park**
Workshop on Agentic Software Engineering (**KDD 2026 Workshop**)
- [W19] Why and When Visual Token Pruning Fails? A Study on Relevant Visual Information Shift in MLLMs Decoding
Jiwan Kim, Kibum Kim, Wonjoong Kim, Byung-Kwan Lee, **Chanyoung Park**
The Fourth International Workshop on Resource-Efficient Learning for Knowledge Discovery (**KDD 2026 Workshop**)
- [W18] Revisiting CF-Integrated LLM Recommenders through Dimensional Collapse
Jimin Seo, HongSeok Kang, Sein Kim, Jiwan Kim, **Chanyoung Park**
Geometric Space, Architecture and Learning Objectives for Large Pre-Trained Models (**KDD 2026 Workshop**)
- [W17] Physics-Embedded Graph Neural Operator for Interaction-Controlled Colloidal Aggregation
Yongjoon Choe, Sungwon Kim, Susan E. Burns, **Chanyoung Park**
Workshop on Reliable Scientific Foundation Models (**KDD 2026 Workshop**)
- [W16] EqGINO: Equivariant Geometry-Informed Fourier Neural Operators for 3D PDEs
Sungwon Kim, Juho Song, Seungmin Shin, Guimok Cho, Sangkook Kim, **Chanyoung Park**
ICLR 2026 Workshop on AI and Partial Differential Equations (Oral) (ICLR 2026 Workshop)
- [W15] Self-EvolveRec: Self-Evolving Recommender Systems with LLM-based Directional Feedback
Sein Kim, Sangwu Park, HongSeok Kang, Wonjoong Kim, Jimin Seo, Yeonjun In, Kanghoon Yoon, **Chanyoung Park**
ICLR 2026 Workshop on AI with Recursive Self-Improvement (RSI 2026) (ICLR 2026 Workshop)
- [W14] MSP-LLM: A Unified Large Language Model Framework for Complete Material Synthesis Planning
Heewoong Noh, Gyoung S. Na, Namkyeong Lee, **Chanyoung Park**
AI4Mat: AI for Accelerated Materials Design (ICLR 2026 Workshop)
- [W13] Synergistic Multi-Task Learning for Electronic Density of States Prediction
Kunmin Jang, Dongik Park, Jaewon Bae, **Chanyoung Park**
AI4Mat: AI for Accelerated Materials Design (ICLR 2026 Workshop)
- [W12] Learning k-Resolved Electronic Structure via Soft Energy Occupancy Prediction
Dongik Park, Kunmin Jang, Jaewon Bae, Dongin Kim, **Chanyoung Park**
AI4Mat: AI for Accelerated Materials Design (ICLR 2026 Workshop)
- [W11] Learning Adaptive Perturbation-Conditioned Contexts for Robust Transcriptional Response Prediction
Yinhua Piao, Hyomin Kim, Seonghwan Kim, Yunhak Oh, Junhyeok Jeon, Sang-Yeon Hwang, Jaechang Lim, Woo Youn Kim, **Chanyoung Park**, Sungsoo Ahn
ICLR 2026 Workshop on Generative and Experimental Perspectives for Biomolecular Design (ICLR 2026 Workshop)
- [W10] RAG-Enhanced Collaborative LLM Agents for Drug Discovery
Namkyeong Lee, Edward De Brouwer, Ehsan Hajiramezanali, **Chanyoung Park**, Gabriele Scalia
MLGenX Workshop at ICLR 2025 (MLGenX 2025 Spotlight)

- [W9] 3D Interaction Geometric Pre-training for Molecular Relational Learning
Namkyeong Lee, Yunhak Oh, Heewoong Noh, Gyoung S. Na, Tianfan Fu, **Chanyoung Park**
NeurIPS 2024 Workshop on AI for New Drug Modalities (AIDrugX)
- [W8] Subgraph Federated Learning for Local Generalization
Sungwon Kim, Yoonho Lee, Yunhak Oh, Namkyeong Lee, Sukwon Yun, Junseok Lee, Sein Kim, Carl Yang, **Chanyoung Park**
KDD 2024 Workshop on Human-Interpretable AI (Oral) (Best Paper Award)
- [W7] Interpretable Graph Model with Prototype-Based Graph Information Bottleneck
Sangwoo Seo, Sungwon Kim, **Chanyoung Park**
KDD 2024 Workshop on Human-Interpretable AI (Oral) (Best Paper Award)
- [W6] Stoichiometry Representation Learning with Polymorphic Crystal Structures
Namkyeong Lee, Heewoong Noh, Gyoung S. Na, Tianfan Fu, Jimeng Sun, **Chanyoung Park**
NeurIPS 2023 Workshop on AI for Scientific Discovery: From Theory to Practice (AI4Science)
- [W5] Electron-Derived Molecular Representation Learning for Real-World Molecular Physics
Gyoung S. Na, **Chanyoung Park**
NeurIPS 2023 Workshop on AI for Scientific Discovery: From Theory to Practice (AI4Science)
- [W4] Deep single-cell RNA-seq data clustering with graph prototypical contrastive learning
Junseok Lee, Sungwon Kim, Dongmin Hyun, Namkyeong Lee, Yejin Kim, **Chanyoung Park**
ICML 2023 Workshop on Computational Biology
- [W3] Denoising single-cell RNA-seq data using Feature Propagation
Sukwon Yun, Junseok Lee, **Chanyoung Park**
ICML 2023 Workshop on Computational Biology (Oral) (Best Paper Award)
- [W2] Predicting Density of States via Multi-modal Transformer
Namkyeong Lee, Heewoong Noh, Sungwon Kim, Dongmin Hyun, Gyoung S. Na, **Chanyoung Park**
ICLR 2023 Workshop on Machine Learning for Materials (ML4Materials)
- [W1] Shift-Robust Node Classification via Graph Adversarial Clustering
Qi Zhu, Chao Zhang, **Chanyoung Park**, Carl Yang, Jiawei Han
NeurIPS 2022 GLFrontiers Workshop

PATENT

- [P1] Method for context-aware recommendation by considering contextual information of document and apparatus for the same
Donghyun Kim, **Chanyoung Park**, Jinoh Oh, Hwanjo Yu
No. 10-18771610000, issued July 2018.

PREPRINT

- [A1] Adaptive Graph Rewiring to Mitigate Over-Squashing in Mesh-Based GNNs for Fluid Dynamics Simulations
Sangwoo Seo, Hyunsung Kim, Jiwan Kim, **Chanyoung Park**

TEACHING

Instructor

- IE343: Statistical Machine Learning, KAIST 2021S, 2022S, 2023S, 2024S, 2026S
- DS535 (KSE801): Recommender System and Machine Learning on Graphs, KAIST 2021F, 2022F, 2023F, 2024F
- CoE202: Basics of Artificial Intelligence, KAIST 2021F
- DS503: Machine Learning for Data Science (Deep Learning), KAIST 2022S, 2023S, 2024S

Teaching Assistant

- CSED352: Data Communications, POSTECH 2014F
- CSED312: Operating System, POSTECH 2016F

PROFESSIONAL SERVICES

AREA CHAIR / SENIOR PROGRAM COMMITTEE

- The International Conference on Machine Learning (ICML) [2025-present]
- The International Conference on Learning Representations (ICLR) [2025-present]
- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) [2025-present]
- Conference on Neural Information Processing Systems (NeurIPS) [2024-present]
- AAAI Conference on Artificial Intelligence (AAAI) [2026-present]
- ICML 2024 AI4Science Workshop [2024-present]

- NeurIPS Workshop (AI for Science: from Theory to Practice) [2023-present]

EVENT ORGANIZATIONS

- Organizer, KDD'26 Workshop on SciSoc Agents & LLMs [2026]
- General Chair, The ACM International Conference on Information and Knowledge Management (CIKM) [2025]
- Proceedings Chair, The ACM International Conference on Information and Knowledge Management (CIKM) [2023]

JOURNAL EDITORSHIP

- Action Editor in Transactions on Machine Learning Research ([TMLR](#))
- Frontiers in Artificial Intelligence ([Graph representation learning: Methods and Applications](#))
- Frontiers in Big Data ([Reviews in Big Data Networks: 2022](#))